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2 Personalized AI assistants could either promote or hinder the
3 emergence of collective intelligence

4
5 Jaeseung Kim¹ Sang Woo Park^{1,2,*}

6 **1** School of Biological Sciences, Seoul National University, Seoul, Korea

7 **2** Institute for Data Innovation in Science, Seoul National University, Seoul, Korea

8 *Corresponding author: sangwoopark@snu.ac.kr

9 **Abstract**

10 Advances in artificial intelligence (AI) have enabled the wide use of personalized
11 assistance for cognitive tasks, which enhanced individual performance on various
12 fields. However, the impact of AI assistance on collective intelligence remains poorly
13 understood. AI assistance may undermine the emergence of collective intelligence by
14 reducing the individual diversity or introducing structured biases into a population.
15 Here, we develop a human-AI collective prediction model mediated by social learning
16 based on incentive structure by extending the existing works on collective intelligence.
17 Our model uncovers that emergence of collective intelligence is depend on how human
18 and AI interact. When AI works as if it is omniscient and gives assistance regardless
19 of users' interests, collective intelligence will be promoted. However, when AI works
20 as a chat-bot and gives assistance only based on users' interests, collective intelligence
21 could be hindered due to diversity loss and overconfidence on AI. We showed that,
22 to promote collective intelligence, penalizing or incentivizing AI usage is not reliable.
23 Instead, elaborating more diversity-aware incentives could be the key to promote and
24 maintain the collective intelligence.

Introduction

Collective intelligence (CI) describes a phenomenon in which a group’s ability to solve problems, make decisions, or adapt to changing environments exceeds each member’s ability [1, 2, 3, 4, 5]. This phenomenon, also referred to as wisdom of crowds, has thus served as a central mechanism underlying the success of diverse organizations, ranging from animal groups [6, 7, 8] and human societies [2, 9] to machine-learning systems [10, 11]. Understanding the conditions that promote and maintain CI therefore remains a fundamental question across social sciences [12], evolutionary biology [13], and computer science [14]. Addressing this question has become particularly timely as artificial intelligence (AI) has begun to reshape how individuals acquire information, make decisions, and contribute to collective outcomes [15].

The recent emergence of generative AIs and large language models (LLMs) has changed the way individuals engage in various cognitive tasks, ranging from simple everyday tasks to more complex tasks, including scientific discovery [16] and creative writing [17]. In particular, while the use of generative AI models, such as ChatGPT, has been rapidly increasing [18], the potential long-term impact of personal AI assistants on human society, especially the CI, remains uncertain. Some argue that AI can promote CI by enhancing individual expertise [19], and opinion aggregation [20]. Conversely, empirical evidence indicate that the use of AIs can limit the diversity of opinions [17, 21, 22], a key requirement for the emergence of CI [23, 24, 25].

Our ability to predict the long-term impact of personal AI assistants on human CI is further hampered by the apparent discrepancy in the incentives that shape AI uses. On one hand, many industrial sectors are pushing towards increased AI usages, where the number of “tokens” used is often perceived as a proxy for productivity [26, 27]. On the other hand, many academic institutions appear to discourage, and even penalize, AI uses, reflecting concerns about plagiarism, misinformation, and limited independence [28, 29]. Such discrepancy raises a question about whether directly incentivizing or penalizing AI usage will benefit or harm human CI. These observations further highlight the importance of identifying a robust incentive structure that can promote CI in the presence of personal AI assistants. Such incentive structure should seek to retain the benefits of AIs and prevent potential risk, such as preserving the diversity of individual opinions.

To address these questions, we extend previous approaches for modeling CI, where each individual makes independent predictions for the outcome from partial observations of underlying causal factors [12, 13, 30]. In particular, we allow each individual to independently consult with personal AI assistants and revise their predictions accordingly. The resulting model allows us to investigate the robustness of human CIs under different assumptions about the AI model’s ability to make predictions as well as incentive structures that are associated with the prediction task. In doing so, we assess the impact of directly incentivizing or penalizing AI uses on CI and further suggest a balanced incentive structure that can maintain CI in the presence of AI.

66 Main

67 Model

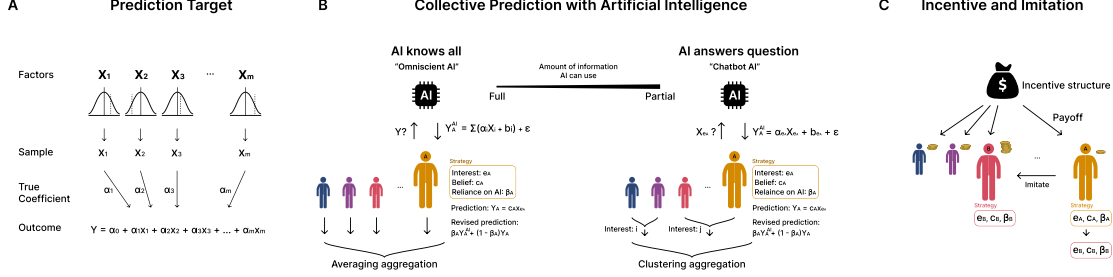


Figure 1: A schematic diagram of the model for individual and collective prediction in the presence of personal AI assistants. [SWP: Need brief description.]

68 We begin by extending previously established models of CI, in which a group of
 69 players participates in a prediction task (Figure 1): each player makes an individual
 70 prediction based on partial observations of the underlying causal factors, and the
 71 group forms a collective prediction by aggregating these individual predictions [30,
 72 12, 13]. Specifically, the prediction target (Y) is modeled as a linear combination of
 73 M causal factors x_i , with fixed coefficients α_i : $Y = \sum_{i=1}^M \alpha_i x_i + \alpha_0$, where each factor
 74 is drawn from independent normal distribution (Figure 1A). Then, each individual
 75 A observes a single factor of their interest x_{e_A} and makes a prediction based on
 76 their own belief c_A : $Y_A = c_A x_{e_A}$ (Figure 1B). We assume each individual can then
 77 independently consult their AI assistants and revise their prediction based on their
 78 reliance on AI β_A : $Y_A^{re} = \beta_A Y_A^{AI} + (1 - \beta_A) Y_A$ (Figure 1B). This revised prediction
 79 is aggregated to form collective prediction. [SWP: I think we need to say averaging
 80 and clustering aggregation here?]

81 To model AI, we assume that AI has already been trained on similar tasks and is
 82 able to make predictions based on partial or full observations of causal factors (Figure
 83 1B). Every time a player consults AI, it will make a new prediction Y_A^{AI} based on
 84 its internal parameters, which can include factor-specific bias (b_i) and stochastic
 85 error ϵ . Here, we consider two extreme types of AI response, which differ in the
 86 amount of information AI uses when making predictions. In one extreme, the AI
 87 can observe all causal factors and use all information to make predictions, regardless
 88 of the user's interest: $Y_A^{AI} = \sum_{i=0}^M (\alpha_i x_i + b_i) + \epsilon$. In the other extreme, AI works
 89 like a chatbot, where the AI predicts the outcome based only on the user's interest:
 90 $Y_A^{AI} = \alpha_{e_A} x_{e_A} + b_{e_A} + \epsilon$. Hereafter, we refer to these two types as 'Omniscient AI'
 91 and 'Chatbot AI', respectively.

92 Each generation consists of multiple round of predictions, and each player is re-
 93 warded a payoff according to the incentive structure (Figure 1C). For simplicity,

we assume that a sufficiently large number of prediction rounds occurs within each generation, allowing us to use the expected payoff value rather than computing the payoffs for every round. We consider two incentive structures from previous work, “feedback” and “niche-expert”, which have been shown to promote near-perfect collective accuracy in the absence of AI [13]. The feedback structure gives higher payoff to individuals who can push the collective prediction toward the true target, regardless of their individual predictive accuracy. [SWP: *maybe we need equations...*] In contrast, the niche-expert structure gives higher payoff to individuals who have both better predictive accuracy and a minority interest. [SWP: *maybe we need equations... but also say that this preserves diversity*] We note collective intelligence fails to emerge under the combination of averaging aggregation with niche-expert structure in the absence of AI [13].

Through time, the group evolves by allowing each individual to imitate peers who have received higher payoffs (Figure 1C). Specifically, after each generation, two individuals are randomly sampled, and one imitates all three strategies (the factor of interest, factor belief, and reliance on AI) of the other with a probability determined by the difference in their expected payoffs. Individual predictions are aggregated into a collective prediction $\hat{Y}$ by simple averaging when using Omniscient AI and by clustering when using Chatbot AI. Collective accuracy is defined as the reduction in prediction error achieved by the collective prediction relative to a baseline prediction α_0 . Hereafter, we refer to the near-perfect collective accuracy as accurate CI.

This model now allows us to explore the impact of AI on CI depending on the mode of AI and the incentive structure. Here, we begin by comparing the emergence of CI with and without AI and subsequently investigate a better incentive structure that can promote CI in the presence of AI.

AI can either promote or prevent the emergence of CI

To understand the impact of AI, we first compare collective dynamics with and without AI. We find that, except when Omniscient AI is used under the feedback structure, a group with AI fails to achieve accurate CI (Fig. 2). However, the effect of AI differs depending on the type of AI.

When Omniscient AI assists players, the group becomes equally or more accurate than the group without AI. Under the feedback structure, a group achieves accurate CI regardless of the assistance of AI (Fig. 2A). However, under the niche-expert structure, the group without AI is no better than the baseline prediction. Although Omniscient AI cannot help players achieve accurate CI, the group’s prediction with AI is more accurate than both that of the group without AI and the AI itself (Fig. 2B). These results are robust across different levels of AI accuracy and can be shown mathematically by assuming a sufficiently large group of players (Fig. S2).

When Chatbot AI assists players, the group becomes less accurate than the group without AI. Although both incentive structures can achieve accurate CI without AI, the assistance of Chatbot AI hinders group performance and prevents accurate CI

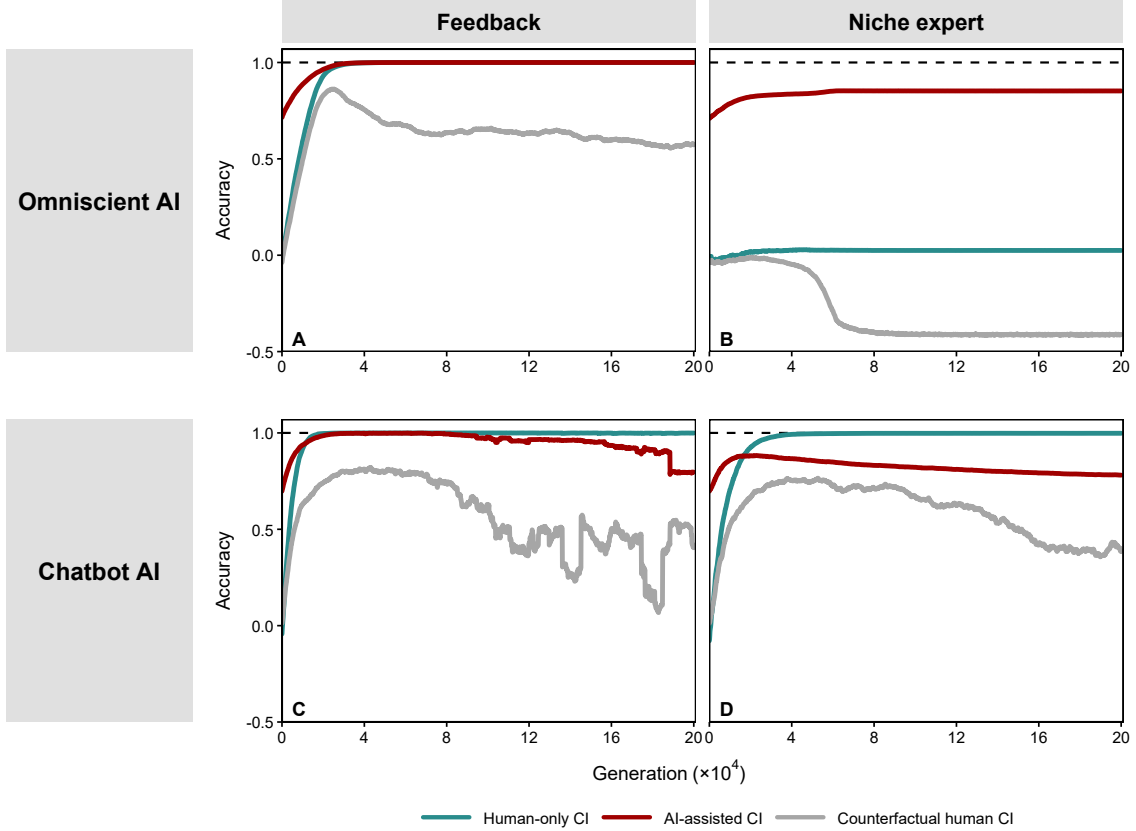


Figure 2: **Need caption.** [SWP: *Need brief description.*]

from emerging (Fig. 2C, D). This occurs through different mechanisms depending on the incentive structure. Under the feedback structure, Chatbot AI leads to a loss of interest diversity, which eventually makes it impossible for the group to account for variation in some causal factors. This leads to an increase in collective variance and stochastic decrease in collective accuracy (Fig. S1). The loss of interest diversity may arise even without AI after sufficiently many generations in a finite population. However, Chatbot AI makes the group more vulnerable to the diversity loss. Under the niche-expert structure, the group becomes overly reliant on AI, causing the collective prediction to inherit the AI’s bias (Fig. S1). This leads to an increase in collective bias and a decrease in collective accuracy. The niche-expert structure may also drive reliance on AI toward zero, depending on the relationship between the true coefficient and AI bias (Fig. S2). In this case, the group completely excludes AI assistance and builds CI on its own, which substantially delays the emergence of accurate CI.

To investigate whether the group can retain its predictive ability when AI assistance suddenly becomes unavailable at a given generation, we measure counterfactual collective accuracy. This measure follows the same evolutionary trajectory

as the simulation with AI, but accuracy is computed using the players' unrevised predictions. We find that counterfactual collective accuracy is always lower than the collective accuracy of groups, regardless of the AI mode or incentive structure.

In summary, AI assistance may promote CI when it is omniscient, whereas AI operating with limited information may prevent the emergence of accurate CI. These findings underscore the need for incentive structures that can mitigate the potential risks associated with AI assistance.

Directly incentivizing or penalizing AI use can additionally harm CI

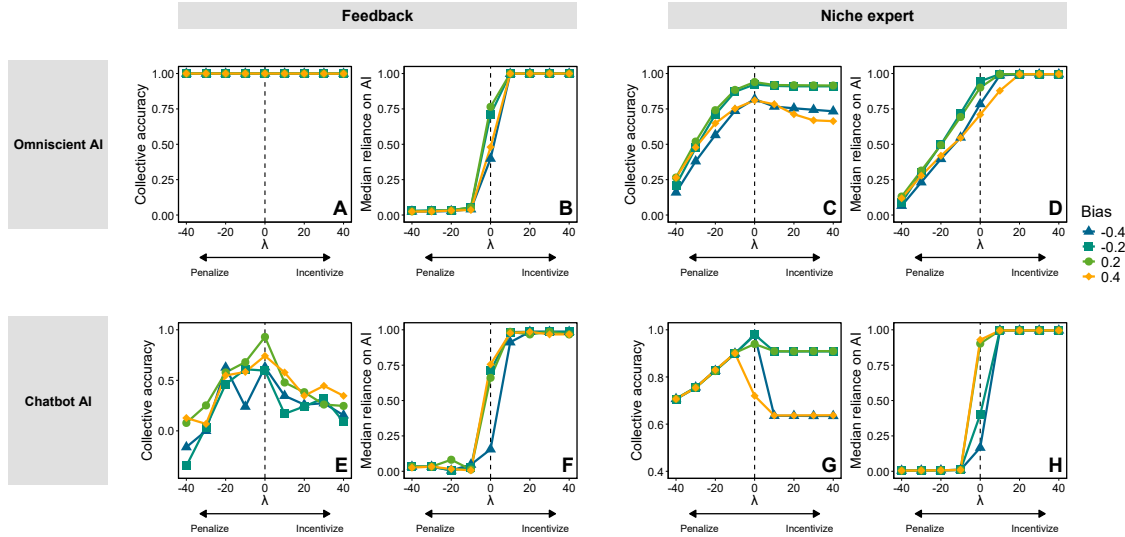


Figure 3: Need caption. [SWP: Need brief description.]

Given the strong influence of AI on collective outcomes, a straightforward intervention is to directly incentivize or penalize the use of AI. To investigate whether such direct incentives or penalties help emergence of CI, we modify the two incentive structures by adding a payoff term proportional to individual reliance on AI:

$$\text{Feedback/Niche-expert} + \lambda\beta_A \quad (1)$$

where λ represent the strength of incentive ($\lambda > 0$) or penalty ($\lambda < 0$).

Directly incentivizing or penalizing AI use further reduce the collective accuracy (Fig. 3C, E, G), except when Omniscient AI is used under the feedback structure (Fig. 3A). This is because incentivizing or penalizing AI usage pushes the group to always ($\beta = 1$) or never ($\beta = 0$) rely on AI, respectively (Fig. 3B, D, F, H). In some cases, when AI bias is relatively large and the penalty is weak, penalizing AI use can achieve higher collective accuracy than the original incentive structure. However, in most cases, penalizing AI use imposes strong selection against AI reliance, pushing

173 the group toward lower stationary collective accuracy. Although the group with
 174 Chatbot AI under feedback structure shows stochastic breakdown of CI which doesn't
 175 result in stationary collective accuracy, incentivizing or penalizing AI use further
 176 strengthens the breakdown (Fig. 3E). This result suggests that simply incentivizing
 177 or penalizing the AI use may exacerbate rather than resolving the problem of AI.

178 **Balanced incentives mitigate AI-induced failures of CI**

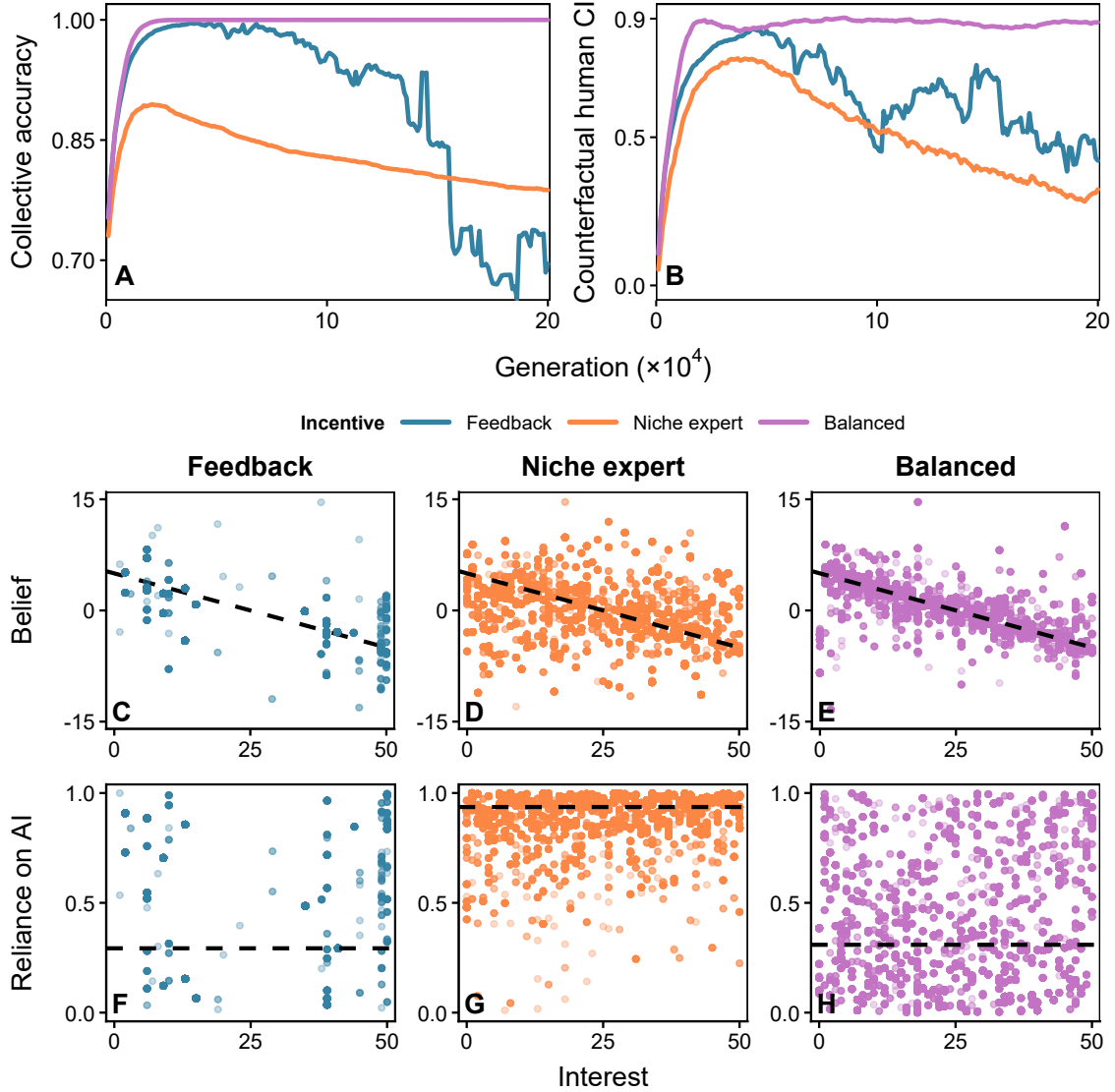


Figure 4: **Need caption.** [SWP: *Need brief description.*]

179 Neither the original incentive structures nor simple interventions targeting AI
 180 use are sufficient to promote CI. What incentive structure, then, would work in the

181 presence of AI? Focusing on the Chatbot AI, in which proposed incentive structures
182 fail to promote accurate CI, we investigate whether a better structure can overcome
183 the risks introduced by AI. Here, we propose a “balanced” structure, defined as the
184 simple average of the feedback and niche expert incentives.

185 Core idea behind the balanced structure is to maintain interest diversity while
186 preventing the collective bias of the group. The feedback structure is prone to in-
187 terest diversity loss, whereas can strongly prevent collective bias by incentivizing
188 players whose predictions lie opposite to the collective prediction. By contrast, the
189 niche-expert structure can maintain interest diversity because it directly incentivizes
190 minority interests. However it cannot prevent collective bias because, within the
191 same factor of interest, individuals are rewarded only according to their individual
192 predictive performance, which homogenize the predictions.

193 By integrating both structure, the balanced structure can promote accurate CI by
194 maintaining both interest diversity and preventing collective bias (Fig. 4, S3). Com-
195 pare to the feedback and niche-expert, players are both evenly distributed through
196 factors, and beliefs are close to the real coefficient (Fig. 4). Moreover, counterfac-
197 tual collective accuracy maintained high compare to both feedback and niche-expert.
198 These patterns are driven by the emergent property of the balanced structure, in
199 which group increases its reliance on AI as the AI becomes more accurate, demon-
200 strating appropriately calibrated AI use.

201 Discussion

202 In this paper, we study how AI and incentive structures affect the emergence of CI.
203 Our model suggests that the effect of AI depends on the type of AI used by the
204 group. AI that can incorporate all relevant causal factors may improve group perfor-
205 mance, whereas AI that relies on users to gather information about the causal factors
206 may impair it. By examining different variants of incentive structures, we further
207 show that direct interventions targeting AI use generally harm group performance.
208 Instead, we suggest balanced structure which can promote CI while also generating
209 appropriately calibrated AI use as an emergent property.

210 Although Omniscient AI can produce group performance that is equal to or bet-
211 ter than that achieved without AI, such AI may be difficult to realize in practice
212 because it needs to be able to observe causal factors independently. Achieving this
213 capability would require hardwares, such as sensors, as well as autonomous data ac-
214 quisition capabilities. For this reason, most AI services currently available are better
215 represented by the Chatbot AI in our model, which highlights the potential risks of
216 AI use for CI.

217 In our model, individuals revise their predictions based on the AI’s assistance. It
218 is therefore unsurprising that AI assistance reduces the diversity of individual pre-
219 dictions, a pattern that has also been observed empirically. However, a reduction in
220 prediction diversity does not necessarily lead to the collapse of CI. The principal risk

arises when the incentive structure fails to guide the group toward an appropriate set of strategies. The feedback structure leads to the breakdown of CI because it recruits more players to more important factors, which have larger coefficients. This characteristic creates a risk of interest-diversity loss in finite populations, a situation in which no player remains interested in certain factors because most players shift toward relatively important factors. Without AI, this occurs when initial individual predictions are not sufficiently diverse or when there is no source of novel ideas, such as mutation. Revising individual predictions with AI assistance exacerbates this problem by pulling individuals’ predictions toward the AI’s prediction, thereby reducing prediction diversity even under the same belief distribution. The niche-expert structure hinders CI because it favors extreme levels of reliance on AI. Because individuals within each interest group are affected only by their own performance, incentives based on individual performance reach their maximum at either full reliance on AI or no reliance at all. This extreme reliance results in suboptimal collective accuracy or substantially delays the emergence of CI.

Because AI already generates substantial benefits as well as costs, organizations increasingly seek to improve collective performance by regulating AI use. Such interventions may take direct forms, including AI-use policies or token-based leaderboards, or may operate indirectly through other incentives, such as productivity rewards or the social stigma associated with AI use. However, our model suggests that directly incentivizing or penalizing AI use is unlikely to induce appropriately calibrated reliance on AI or to promote CI. Incentivizing AI use could lead to overreliance, causing groups to propagate AI biases, whereas penalizing AI use could lead to underreliance, preventing groups from using AI even when it performs well in the relevant domain. Our result instead shows that appropriate AI use and accurate CI can be promoted by focusing on the classic properties of groups that have been emphasized for decades—maintaining diversity and preventing collective bias—rather than on AI use itself.

The consistently limited counterfactual collective accuracy observed across AI types and incentive structures suggests that the presence of AI can degrade human group’s own performance, consistent with empirical evidence. This may reduce collective performance when AI assistance becomes unavailable because of resource constraints, system failure, or model collapse, forcing the group to rebuild CI through further adaptation. The higher counterfactual collective accuracy achieved under the balanced structure indicates that an appropriate incentive structure can mitigate the risk of CI breakdown following the loss of AI assistance.

Our model and analysis have several limitations. First, the model considers only social learning among humans and treats the influence of AI as unidirectional. In reality, humans can learn from AI, while AI systems can also be further trained on human-generated data. Such bidirectional adaptation may further reduce group performance by increasing the similarity between human and AI predictions and thereby weakening their complementarity. Empirical studies of human–AI collaboration likewise suggest that stronger correlations between human and AI predictions

can diminish the benefits of collaboration. Second, we model AI errors using factor-specific additive biases and random error. Previous studies of AI bias and error suggest that additive biases may arise from skewed training data, or systematic calibration. However, AI errors may also exhibit more complex structures, including biased estimates of factor effects and correlations across prediction errors. Further empirical research is therefore needed to characterize the structure of AI bias and error, together with modeling approaches that can incorporate these more complex patterns. Finally, to retain the tractability of established CI models, we assume that the prediction target is a linear combination of causal factors, which does not hold for many real-world tasks. In particular, many machine learning methods used in AI systems are capable of capturing nonlinear relationships. Future research should therefore investigate how nonlinear relationships between causal factors and prediction targets affect the interaction between AI assistance and the emergence of CI.

Methods

Prediction target

Following previous models of CI, we model the prediction target as a linear combination of M independent causal factors:

$$Y = \alpha_0 + \alpha_1 X_1 + \cdots + \alpha_M X_M, \quad (2)$$

where the fixed coefficient α_i determines the magnitude and direction of the effect of factor i , and α_0 is a global intercept. Each causal factor X_i is modeled as an independent Gaussian random variable with mean zero and standard deviation σ_i .

Players

We consider a well-mixed population N consisting of n players engaged in a novel prediction task. In the absence of AI assistance, the strategy of player A is characterized by two variables: interest e_A and belief c_A . Player A observes the causal factor X_{e_A} and predicts the target by multiplying the observed factor value by their belief:

$$Y_A = c_A X_{e_A}, \quad (3)$$

where e_A can take an integer value from 0 to M . When $e_A = 0$, the player does not observe any causal factor and instead makes a constant prediction based solely on their belief. We refer to this as a stubborn strategy because the player's prediction does not respond to the realized values of the causal factors. For notational convenience, we assume that $X_0 = 1$.

295 When the group is assisted by AI, each player’s strategy additionally includes
 296 reliance on AI, denoted by β_A . We assume that each player consults the AI inde-
 297 pendently and revises their prediction by taking a weighted average of their own
 298 prediction and the AI-generated prediction:

$$Y_A^{\text{re}} = \beta_A Y_A^{\text{AI}} + (1 - \beta_A) Y_A, \quad (4)$$

299 where β_A ranges from 0 to 1. A value of $\beta_A = 1$ indicates complete reliance on
 300 AI, whereas $\beta_A = 0$ indicates that the player does not use the AI prediction.

301 AI models

302 We assume that the AI model has been trained on similar tasks and is therefore
 303 capable of making predictions based on causal factors. However, because the model
 304 is imperfect, its outputs are subject to systematic additive bias and stochastic error.
 305 The additive bias associated with factor i is denoted by b_i , and the stochastic error
 306 is denoted by ϵ . The AI is neither trained nor updated during the simulation, and its
 307 biases therefore remain fixed throughout the simulation. Although the bias is indexed
 308 by factor, all b_i , including b_0 , are assigned the same value in the main analyses for
 309 simplicity. The model formulation nevertheless allows biases to differ among factors.

310 We assume that the stochastic error follows a Gaussian distribution with mean
 311 zero and standard deviation τ . Each time a player consults the AI, an indepen-
 312 dent error is generated. Errors are therefore independent across players and across
 313 repeated consultations by the same player. Consequently, stochastic AI errors can
 314 be attenuated when independently generated predictions are aggregated across play-
 315 ers. We therefore held τ constant and varied the magnitude of the additive bias to
 316 manipulate the level of AI accuracy across analyses.

317 We consider two extreme types of AI that differ in the amount of information
 318 available for prediction. The first is termed the Omniscient AI, which can observe
 319 all M causal factors and incorporate them into its prediction regardless of the user’s
 320 interest. The prediction generated by the Omniscient AI for player A is expressed
 321 as follows:

$$Y_A^{\text{AI}} = \alpha_0 + b_0 + \alpha_1 X_1 + b_1 + \cdots + \alpha_M X_M + b_M + \epsilon. \quad (5)$$

322 The second is termed the Chatbot AI, which can use only the single causal factor
 323 X_{e_A} associated with the player’s interest. The prediction generated by the Chatbot
 324 AI for player A is expressed as follows:

$$Y_A^{\text{AI}} = \alpha_{e_A} X_{e_A} + b_{e_A} + \epsilon. \quad (6)$$

325 Aggregation rules

326 The revised individual predictions are aggregated to form a collective prediction. We
 327 use two aggregation rules proposed in previous research, depending on the type of

328 AI used by the group.

329 When the group is assisted by the Omniscient AI, the AI component of each
 330 revised prediction contains information about all M factors, even though the indi-
 331 vidual’s own prediction is based on only one factor. Each revised prediction can
 332 therefore be treated as an estimate of the entire target. Accordingly, we calculate
 333 the collective prediction by taking the simple average of the revised individual pre-
 334 dictions:

$$\hat{Y}^{\text{re}} = \frac{Y_1^{\text{re}} + Y_2^{\text{re}} + \cdots + Y_n^{\text{re}}}{n}. \quad (7)$$

335 When the group is assisted by the Chatbot AI, both the player’s initial prediction
 336 and the AI’s prediction are based on the same causal factor. Each revised predic-
 337 tion therefore contains information about only one factor and can be treated as an
 338 estimate of that factor’s contribution to the target. We consequently use clustering
 339 aggregation.

340 Clustering aggregation is performed by grouping individuals according to their
 341 interests and calculating the mean prediction within each group. The collective
 342 prediction is then obtained by summing the mean predictions of all nonempty groups:

$$\hat{Y}^{\text{re}} = \sum_{i:n_i>0} \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} Y_A^{\text{re}}, \quad (8)$$

343 where $\mathcal{A}_i$ is the set of players interested in factor i , and $n_i = |\mathcal{A}_i|$ is the number
 344 of players in that set.

345 When the group is not assisted by AI, collective predictions are computed from the
 346 unrevised individual predictions throughout the simulation. Both simple averaging
 347 and clustering can be used in the absence of AI. For fair comparison, we use the
 348 same aggregation rule as in the corresponding AI-assisted condition.

349 Incentive structures

350 Players receive payoffs determined by an incentive structure based on their strategies.
 351 We consider five incentive structures: feedback structure, niche-expert structure,
 352 direct intervention on AI use of these two structures, and a balanced structure.

353 The feedback structure does not directly reward individuals according to their
 354 own prediction accuracy. Instead, it rewards predictions that are aligned with the
 355 residual error of the collective prediction and therefore have the potential to move
 356 the collective prediction toward the true value. Given player A ’s revised prediction
 357 Y_A^{re} and the collective prediction $\hat{Y}^{\text{re}}$, the payoff is expressed as

$$\pi_A = Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}). \quad (9)$$

358 The niche-expert structure assigns higher payoffs to individuals who make accu-
 359 rate predictions while having an underrepresented interest. Specifically, the squared

360 prediction error of player A is weighted by $\rho_{e_A} = n_{e_A}/n$, the proportion of players
 361 interested in the same factor:

$$\pi_A = -\rho_{e_A}(Y_A^{\text{re}} - Y)^2. \quad (10)$$

362 Previous research has shown that these two incentive structures can produce
 363 nearly perfect CI through social learning, except when the niche-expert incentive is
 364 combined with simple averaging. They are therefore used as baseline structures for
 365 examining how AI assistance affects the emergence of CI.

366 To examine the effects of directly encouraging or discouraging AI use, we mod-
 367 ify the feedback and niche-expert structures by adding a term proportional to the
 368 player’s reliance on AI:

$$\pi_A = Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) + \lambda\beta_A. \quad (11)$$

$$\pi_A = -\rho_{e_A}(Y_A^{\text{re}} - Y)^2 + \lambda\beta_A. \quad (12)$$

369 Here, λ is a weighting coefficient representing the strength of the direct incentive
 370 or penalty applied to AI use. A positive value of λ rewards AI reliance, whereas a
 371 negative value penalizes it. When $\lambda = 0$, no direct incentive or penalty is applied,
 372 and the payoff functions reduce to the original feedback and niche-expert structures.

373 The balanced structure is defined as the arithmetic mean of the feedback and
 374 niche-expert payoff components:

$$\pi_A = \frac{Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) - \rho_{e_A}(Y_A^{\text{re}} - Y)^2}{2}. \quad (13)$$

375 Simulations

376 All simulations were conducted using an agent-based model with 50 causal factors
 377 ($M = 50$) and 10,000 players ($n = 10,000$). The parameters defining the prediction
 378 target were sampled at the beginning of the simulations and subsequently held fixed.

379 The coefficients α_i were sampled from a uniform distribution on $[-5, 5]$, except in
 380 the simulation presented in Figure 4, in which α_i was set to $5 - 0.2i$ for visualization.
 381 The standard deviations σ_i were sampled from a uniform distribution on $[0, 3]$.

382 Players’ initial strategies were also sampled from predefined probability distri-
 383 butions. The distribution of individual belief c_A depended on the aggregation rule.
 384 Under simple averaging, each factor’s contribution is scaled by the proportion of
 385 players interested in that factor. We therefore sampled beliefs from a Gaussian dis-
 386 tribution with mean zero and standard deviation 100, following the scaling used
 387 in previous work [13]. Under clustering aggregation, beliefs were sampled from a
 388 Gaussian distribution with mean zero and standard deviation 5. Because predictions
 389 are averaged separately within each interest group and then summed, no additional
 390 scaling of c_A is required. The initial interest of each player was sampled uniformly

391 from the integers between 0 and 50, and reliance on AI was sampled from a uniform
 392 distribution on $[0, 1]$.

393 We assumed that prediction events occurred on a substantially faster timescale
 394 than social learning. At each generation, we therefore calculated the expected payoff
 395 $\mathbb{E}[\pi_A]$ of each player analytically rather than explicitly sampling prediction events.
 396 The expectation was taken over the distributions of the causal factors $X_1, \dots, X_M$
 397 and the stochastic AI error ϵ , conditional on the current population strategies and
 398 the fixed target parameters. Closed form expression of expected payoffs were derived
 399 for all incentive structures and are presented in the Supplementary Methods.

400 Each generation consisted of an expected payoff calculation followed by one op-
 401 portunity for pairwise social learning. Two distinct players, A and B , were randomly
 402 sampled, and player A imitated all three strategies of player B with probability

$$\frac{1}{1 + \exp(s(\mathbb{E}[\pi_A] - \mathbb{E}[\pi_B]))}, \quad (14)$$

403 where s represents the strength of social learning and was set to 50 in all sim-
 404 ulations. We introduced a mutation rate μ , such that with probability μ , player A
 405 did not imitate player B but instead redrew all strategy components from the same
 406 distributions used to initialize the population. The mutation rate was set to zero in
 407 all simulations except those presented in Supplementary Figure 4.

408 Unless otherwise specified, each simulation was run for 200,000 generations. When
 409 a result was represented by a single summary value rather than as a temporal tra-
 410 jectory, the metric was averaged over the final 10,000 generations.

411 To evaluate the dynamics of collective prediction, we defined six summary met-
 412 rics: collective accuracy, collective bias, collective variance, counterfactual collective
 413 accuracy, interest diversity, and median reliance on AI. At each population state,
 414 expectations and variances were evaluated analytically over the distributions of the
 415 causal factors and AI errors rather than estimated from sampled prediction events.
 416 Collective bias and collective variance constitute the bias-variance decomposition of
 417 the group level collective prediction error.

418 Collective accuracy was normalized relative to the baseline prediction α_0 . A neg-
 419 ative value indicates that the collective prediction performs worse than the baseline.
 420 Counterfactual collective accuracy was evaluated by setting $\beta_A = 0$ while retaining
 421 the evolved interests and beliefs of the AI-assisted population. In calculating interest
 422 diversity, we used the convention that $\rho_i \ln \rho_i = 0$ when $\rho_i = 0$ (Table 1).

423 Data and code availability

424 All data are stored in a publicly available GitHub repository ([https://github.com/](https://github.com/parklab-snu/collective-intelligence-with-AI)
 425 [parklab-snu/collective-intelligence-with-AI](https://github.com/parklab-snu/collective-intelligence-with-AI)).

Table 1: Evaluation metrics used to analyze collective prediction dynamics.

Metric	Definition	Interpretation
Collective accuracy	$1 - \frac{\mathbb{E}[(Y - \hat{Y}^{\text{re}})^2]}{\mathbb{E}[(Y - \alpha_0)^2]}$	Accuracy of the revised collective prediction relative to the baseline prediction α_0 .
Collective bias	$\left(\mathbb{E}[Y - \hat{Y}^{\text{re}}]\right)^2$	Squared expected deviation of the revised collective prediction from the true target.
Collective variance	$\text{Var}(Y - \hat{Y}^{\text{re}})$	Variability of the revised collective prediction error across prediction events.
Counterfactual collective accuracy	$1 - \frac{\mathbb{E}[(Y - \hat{Y})^2]}{\mathbb{E}[(Y - \alpha_0)^2]}$	Accuracy obtained after removing AI use from the current population while retaining its evolved interests and beliefs.
Interest diversity	$\exp\left(-\sum_{i=0}^M \rho_i \ln \rho_i\right)$	Effective number of interests represented in the population.
Median reliance on AI	$\text{median}_{A \in N}(\beta_A)$	Median reliance on AI in the population.

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430 **Contributions**

431 Jaeseung Kim: Conceptualization, Formal Analysis, Investigation, Methodology,
432 Software, Visualization, Writing - Original Draft Preparation, Writing - Review &
433 Editing. Sang Woo Park: Conceptualization, Investigation, Methodology, Supervi-
434 sion, Writing - Original Draft Preparation, Writing - Review & Editing.

435 **Competing Interests**

436 The authors declare no competing interests.

⁴³⁷ **Supplementary Text**

438 **Supplementary Figure**

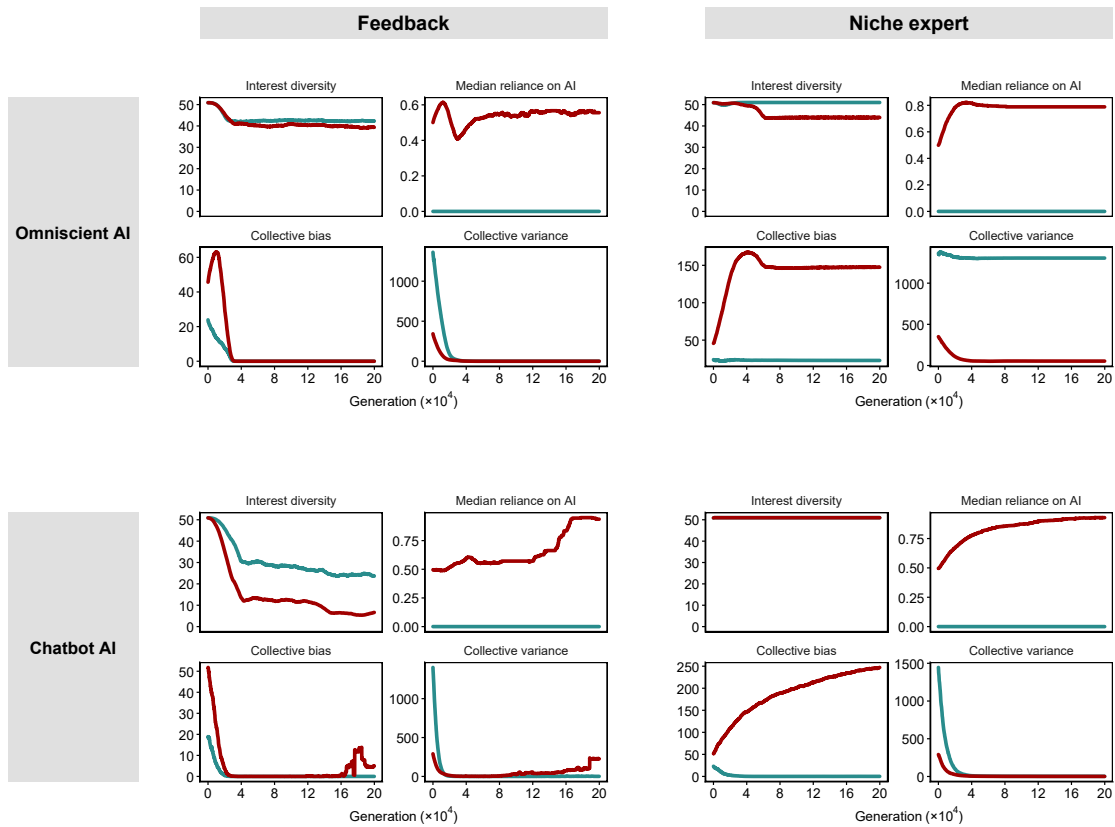


Figure S1: **Caption.** Text.

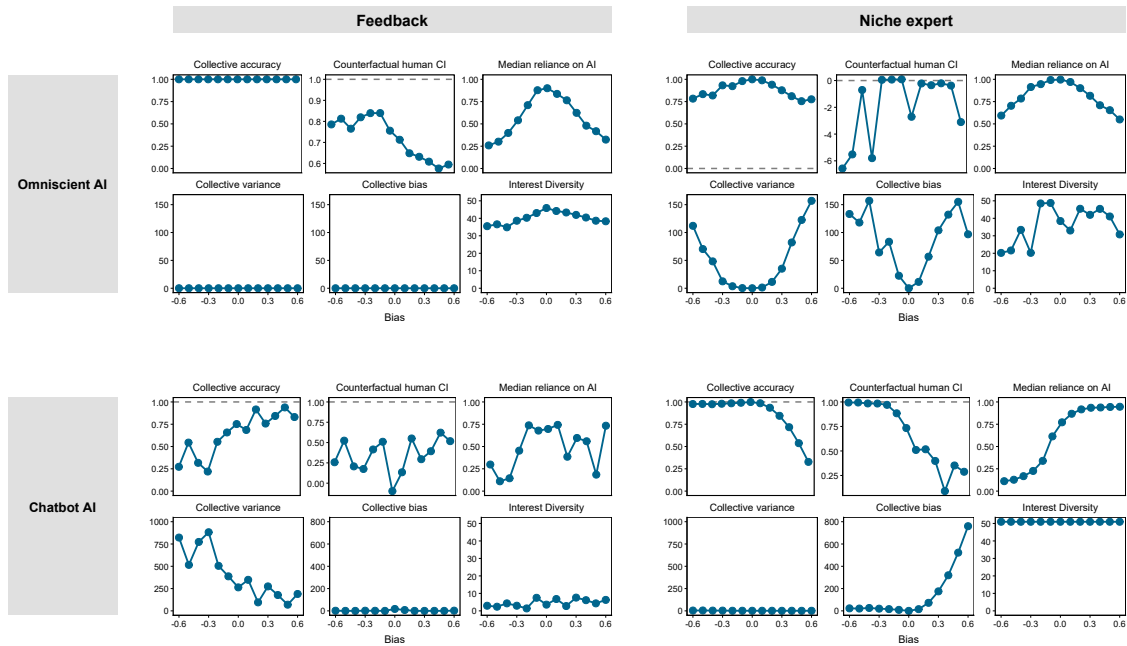


Figure S2: Caption. Text.

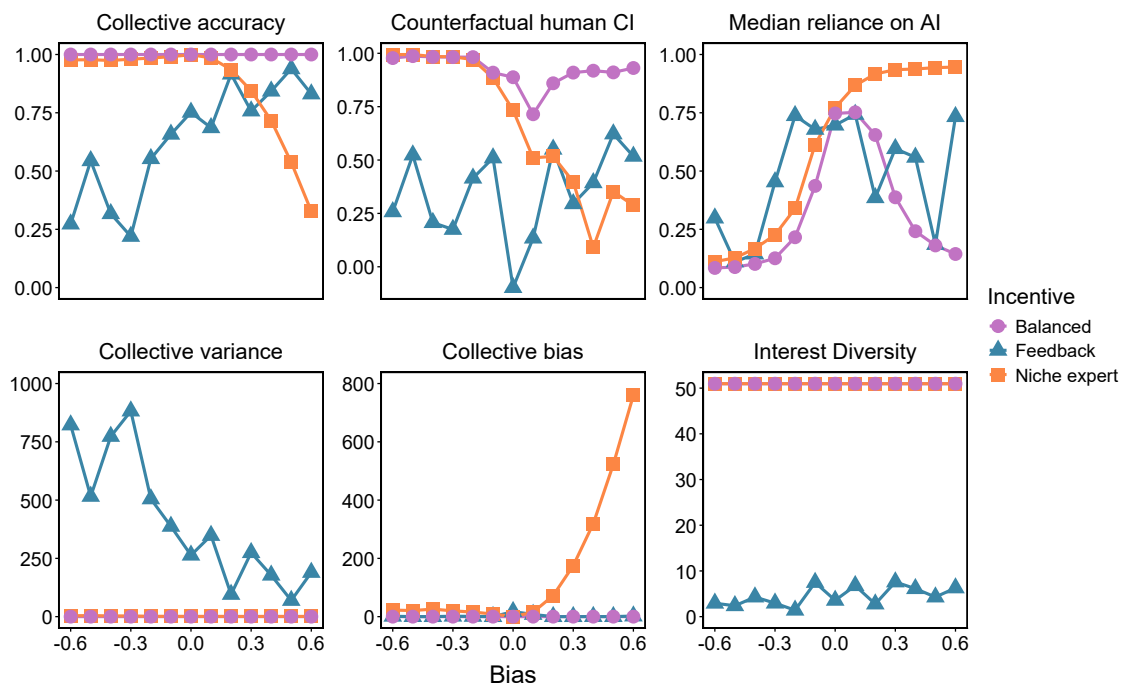


Figure S3: **Caption.** Text.

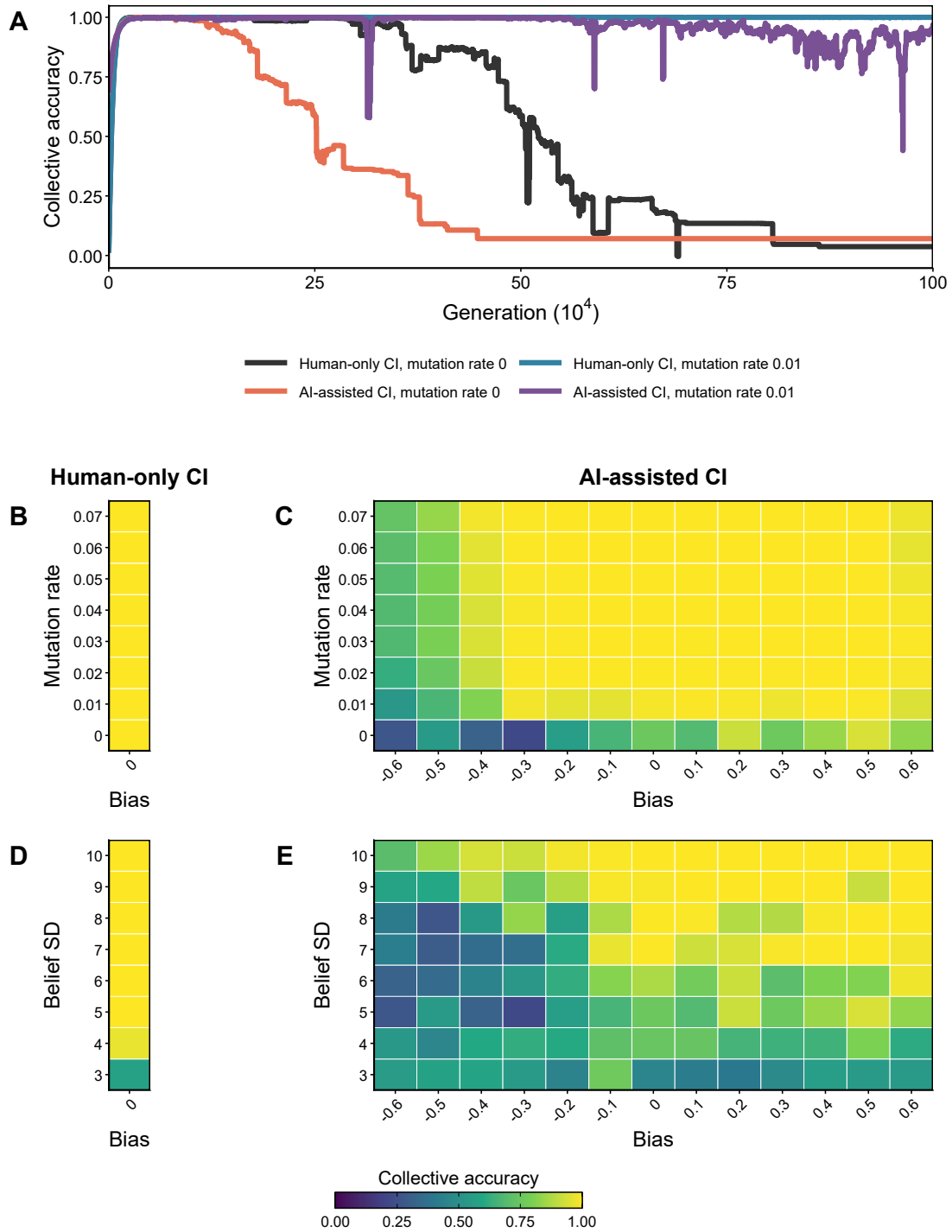


Figure S4: **Caption.** Text.

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